

# Dalton Kutzen

## Curriculum Vitae

[dalton@kutzen.org](mailto:dalton@kutzen.org) · [GitHub.com/dkutzen22](https://github.com/dkutzen22) · [LinkedIn.com/in/daltonkutzen](https://www.linkedin.com/in/daltonkutzen)

## Education

---

**Brigham Young University — B.S. Molecular Biology**

Expected Apr 2028

Provo, Utah

## Experience

---

**Bioinformatics Researcher**

Aug 2026 – Present

[Terooatea Lab, Brigham Young University](#) · Provo, UT

- Developed computational methods and benchmark evaluations for high-dimensional single-cell RNA-seq (scRNA-seq) and pooled Perturb-seq screens (anchor-op).
- Implemented target-grouped nested cross-validation and matched signal-to-noise ratio (SNR) controls in Python to evaluate prediction performance against baseline models.
- Co-authored bioRxiv research on perturbation screen modeling and computational reproducibility.

**Research Assistant**

Sep 2025 – Present

[Grose Lab, Brigham Young University](#) · Provo, UT

- Prepared media, cultured bacteriophages with bacterial hosts, and conducted plaque and spot assays to evaluate phage activity and host interactions.
- Performed bioinformatic analyses to annotate and compare phage genomes, creating the **Phantom** Snakemake pipeline for reproducible annotation.
- Integrated wet lab microbiology techniques with computational genomics to investigate phage diversity, host range, and therapeutic applications.

**IT Service Desk Supervisor**

May 2026 – Present

[College of Life Sciences, Brigham Young University](#) · Provo, UT

- Mentor student technicians in troubleshooting best practices and train new hires on Life Sciences IT infrastructure, tools, and clear user communication.
- Drove internal automation, documentation, and reporting initiatives that reduced repetitive workflows and standardized support delivery college-wide.

**IT Service Desk Technician**

Aug 2025 – May 2026

- Oversaw deployment, security, and maintenance of 1,000+ campus workstations and lab computers, Supporting a community of 5,000+ users

## Publications & Preprints

---

- Fullmer, K. J., **Kutzen, D.**, Terooatea, T. W. (2026). *Additive-input encoding fails operator-level target-held-out prediction on current Perturb-seq screens.* preprint, *bioRxiv*. [[anchor-op](#)]

## Presentations & Posters

---

- Duffy, M. E., **Kutzen, D.**, Grose, J. H. (Apr 2026). *Characterizing Proteus mirabilis Phages for Clinical Applications.* Poster Presentation, American Society for Microbiology, Intermountain Branch Conference, Salt Lake City, UT

## Selected Projects

---

**Phantom — Bacteriophage Analysis Workflow**

Jun 2026 – Present

- Modular Snakemake pipeline for phage genome annotation and downstream comparative analysis.
- Uses isolated Conda environments and declarative workflow rules to make runs reproducible across systems.

- Built a Linux-based environment for code hosting, containers, services, and research project organization.
- Applied infrastructure-as-code and documentation practices to reduce configuration drift.

## Skills

| Area                                   | Tools & Proficiencies                                                                                                                                                          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Computational Biology</b>           | Multi-omics analysis, comparative genomics, Snakemake, Conda, Biopython                                                                                                        |
| <b>Wet Lab &amp; Molecular Biology</b> | Phage and bacterial culturing, plaque & spot assays, host range testing, media preparation, agarose gel electrophoresis                                                        |
| <b>Programming &amp; Data</b>          | Python, R, Bash, SQL, Git, pandas, tidyverse, ggplot2, Quarto                                                                                                                  |
| <b>IT &amp; Systems Infrastructure</b> | IT Service Management (ITSM), Linux server administration, Proxmox, Docker, enterprise endpoint management, networking, project management, technical mentoring, documentation |

## Leadership & Service

### Initiative Lead

Jan 2026 – Present

Leadership and the Scriptures, Brigham Young University · Provo, UT

- Plan and facilitate structured discussions connecting scriptural principles with practical leadership in academic, community, and professional settings.
- Guide participants in developing actionable leadership habits centered on empathy, honoring agency, and fostering personal and interpersonal accountability.

### Counselor and Youth Mentor

May 2025 – Aug 2025

FSY Conferences · United States

- Mentored cohorts of 15–25 youth participants, facilitating daily meetings, leadership workshops, and goal-setting activities.
- Motivated participants to establish actionable personal and academic goals, develop effective study habits, and strengthen character traits.
- Reported regularly to conference leadership regarding participant progress, group dynamics, and logistical improvements.

### Full-Time Volunteer

Mar 2022 – Mar 2024

The Church of Jesus Christ of Latter-day Saints · Wichita, Kansas

- Supervised and trained 200+ volunteers throughout the state of Kansas, enhancing operational productivity and organizational goal execution.
- Provided daily instruction, counseling, and community service to individuals and families to support personal well-being and life skills.

## Honors & Awards

|                                                                                                             |             |
|-------------------------------------------------------------------------------------------------------------|-------------|
| • <b>Sacramento Alumni Association Scholarship</b> , Brigham Young University                               | Winter 2027 |
| • <b>Experiential Learning Grant</b> , Department of Micro and Molecular Biology - Brigham Young University | Fall 2026   |
| • <b>William Bartlett Hebard Scholarship</b> , College of Life Sciences - Brigham Young University          | Fall 2026   |
| • <b>Experiential Learning Grant</b> , Department of Micro and Molecular Biology - Brigham Young University | Spring 2026 |